

Md Hassan

INDIAN INSTITUTE OF TECHNOLOGY, KANPUR

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PROFILE

Computer Science graduate currently pursuing an M.Tech. at IIT Kanpur, with a strong academic foundation in Machine Learning, Artificial Intelligence, Computer Systems, and Networks. My research interests include Large Language Models, Generative AI, multimodal learning, and AI-driven cybersecurity. My recent work focuses on encrypted traffic analysis and intelligent network security, where I designed end-to-end ML pipelines. I seek to pursue doctoral research that advances foundation models and intelligent AI systems through rigorous and impactful research.

EDUCATION

Indian Institute of Technology (IIT) Kanpur

MASTER OF TECHNOLOGY (M.TECH.) [🔗](#)

2026 - Present

Kanpur, India

- Focus: Machine Learning, Generative AI, Photonics, LLM

Maulana Azad National Urdu University (MANUU), Hyderabad

B.TECH. IN COMPUTER SCIENCE · CGPA: 8.96/10 [🔗](#)

2022 - 2026

Hyderabad, India

- Coursework: Machine Learning, Deep Learning, Network Security, Data Structures & Algorithms, Computer Networks, Cryptography, Operating Systems, Database Systems

RESEARCH & TECHNICAL PROJECTS

Multi-Layered VPN Detection & Traffic Deanonimisation

2025 - 2026

LEAD RESEARCHER

[🔗](#) Project page)

- Engineered a hybrid **Deep Packet Inspection (DPI) + Machine Learning** pipeline classifying encrypted VPN tunnels across 5+ protocol types (OpenVPN, WireGuard, Shadowsocks, I2P).
- Achieved **90%+ protocol-level classification accuracy** using flow metadata fusion, statistical fingerprinting, and ensemble feature selection.
- Built high-throughput packet parsing pipeline with Scapy, processing 10,000+ packet flows per session for real-time analysis.

AI-Powered Intrusion Detection System (IDS)

2024

ML ENGINEER

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- Designed and deployed a **real-time ML-based IDS** for high-throughput network threat classification and anomaly detection using Python and Scikit-learn.
- Applied feature engineering on NetFlow/PCAP data to improve detection precision; optimized inference pipeline to **sub-10ms latency** using containerized Docker deployment.
- Benchmarked against signature-based IDS baselines, demonstrating measurable ML uplift across attack categories.

M-ACT: Mirrored Active Collaborative Tunnel

2025

SYSTEMS DEVELOPER

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- Architected a real-time collaborative tunneling platform enabling persistent remote access to local environments via dynamic URL mirroring.
- Engineered secure system-sharing protocols supporting distributed workflows, reducing remote debugging cycle time through synchronized local-to-cloud replication.

Learn2Play: AI-Driven Adaptive Learning System

2024 - 2025

AI DEVELOPER

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- Built an **ML-based adaptive content recommendation engine** using behaviour modelling, real-time engagement signals, and data-driven personalisation.
- Applied reinforcement learning principles to optimize long-term content sequencing; integrated behavioral analytics for continuous model refinement.

INDUSTRY EXPERIENCE

FlyRank AI

July 2026 – Present

ML ENGINEERING INTERN

Remote

- Building and optimizing production **ML models and scalable inference workflows** for AI-driven marketing analytics and content optimization.
- Contributing to end-to-end ML pipelines: data ingestion, feature engineering, model training, serving, and performance monitoring.

India Space Lab

May – June 2026

SPACE TECHNOLOGY INTERN

Remote

- Completed intensive technical training in CanSat/CubeSat systems, Remote Sensing, GIS analysis, Rocketry Science, and Disaster Management protocols.
- Applied systems engineering and data analysis methods to satellite mission design workflows.

Skillship Hyderabad (NGO)

Jan 2023 – Jan 2024

GRAPHICS DESIGNER INTERN

Hyderabad, India

- Maintained consistent visual branding across digital and print assets while collaborating with team members to communicate information effectively.
- Created engaging visual content and presentation materials that enhanced participant engagement during workshops, training sessions, and community initiatives.

SKILLS

AI / ML Machine Learning, Generative AI, Large Language Models (LLMs), Prompt Engineering, Scikit-learn, TensorFlow, XGBoost, Feature Engineering, Classification, Anomaly Detection, Reinforcement Learning

Cybersecurity DPI, VPN, IDS/IPS, Network Forensics, Traffic Analysis, SIEM

Programming Python, C, C++

Data Science & Computing Numpy, Pandas, Jupyter Notebook

Frameworks FastAPI, Scapy, Docker, Git, Wireshark, Nmap, Metasploit, Splunk

Networking TCP/IP, UDP, NetFlow, PCAP, OpenVPN, WireGuard, HTTP/S, DNS, TLS/SSL

SCHOLARSHIPS & ACHIEVEMENTS

- **GATE 2026** – Qualified in Computer Science (CS) and Data Science & AI (DA), IIT Guwahati
- **GATE 2025** – Qualified in Computer Science (CS), IIT Roorkee
- **UGC NET 2025** – Qualified in Computer Science & Applications
- **CGPA 8.96/10** – Top standing, B.Tech. Computer Science, MANUU Hyderabad

LEADERSHIP & SERVICE

Entrepreneurship Cell (E-Cell), MANUU Hyderabad

Aug 2023 – May 2026

FOUNDING MEMBER

Hyderabad, India

- Founding board member of the university's inaugural E-Cell as a Dune & Digit Team, building a student innovation ecosystem supporting AI, cybersecurity, and data-driven startup initiatives.
- Mentored 5+ student teams on technology project development and startup ideation; organized interdisciplinary innovation workshops.

Kanpur, India · Last updated August 2026 · References available on request